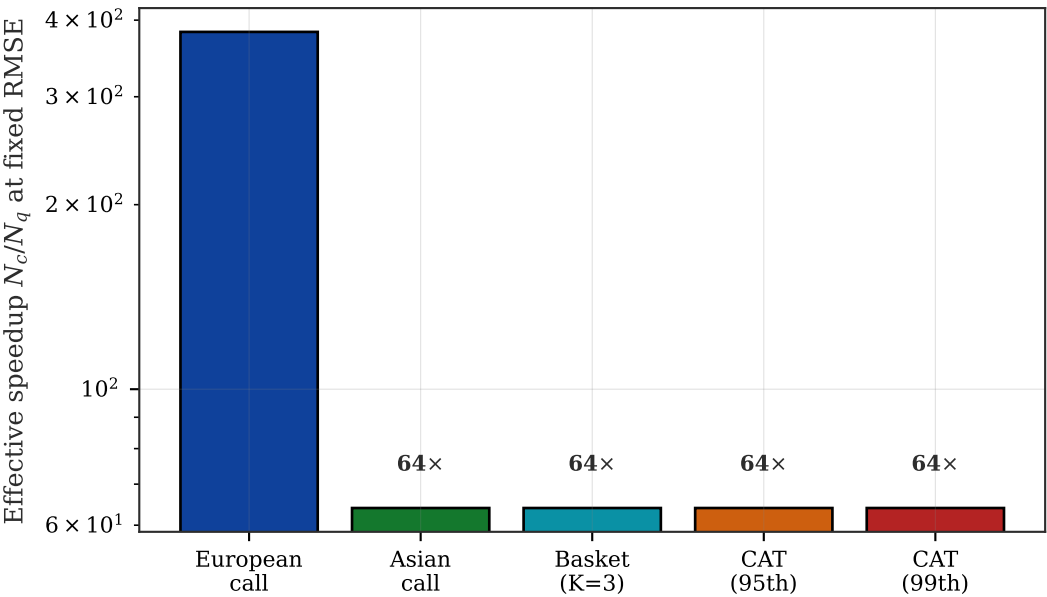
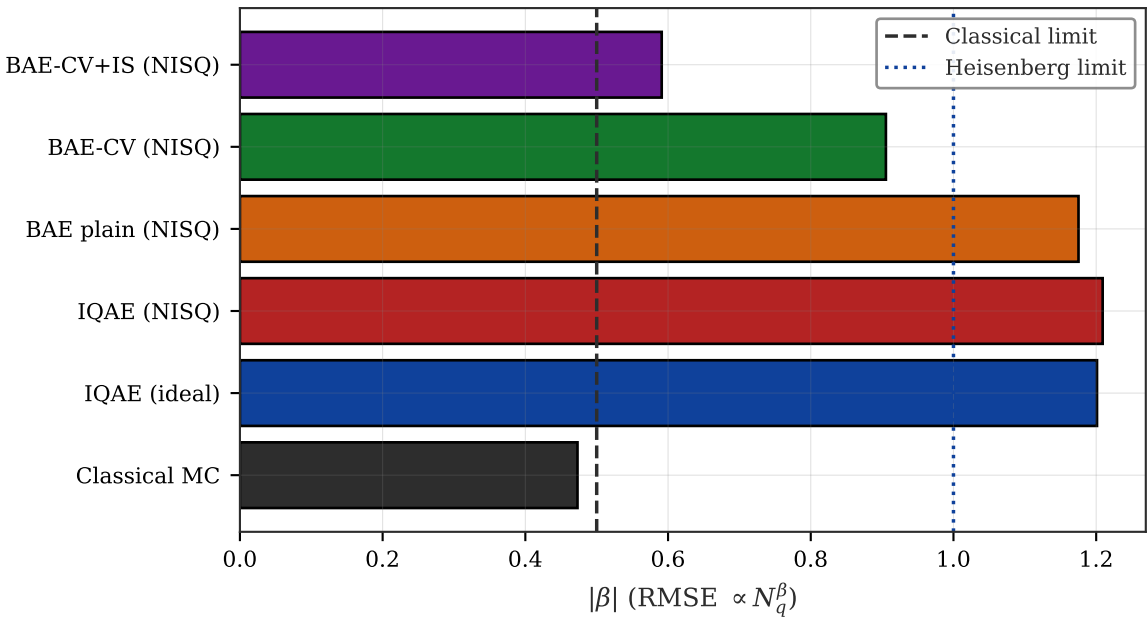


Figure 12 — Summary Dashboard: Hybrid QCMC Results Across All Scenarios

(a) Hybrid BAE-CV+IS Effective Speedup by Scenario



(b) Simulated Convergence Exponents (European Call)



(c) Consolidated QRMF Risk Report

Quantum Risk Management Framework—Key Computed Quantities

European call (BS closed form) : \$10.4506
Asian call (12-step MC, 4,000,000 paths) : \$6.1551
3-asset basket call (Cholesky circuit) : \$8.3019
CAT excess loss, 95th pct (14,784 events) : \$8.05 ($\times 10^5$)
CAT excess loss, 99th pct : \$5.81 ($\times 10^5$)

ρ_{CV} (European, forward control) : 0.9245 ($R_{CV}=6.89\times$)
Dual-CV Asian $\rho^\top R_w^{-1} \rho$: 0.9994
 R_{IS} (European call) : 1.70x
 R_{IS} (CAT, 99th pct.) : 8.89x

Systemic risk: most systemic factor : HML
QUBO ground energy (exact) : -0.3983
QAOA approx. bitstring match to exact : YES
Hybrid QRAE VaR99% error vs. std. QRAE : 3398 vs 16848

(d) Query-Complexity Theorem Factor Decomposition (Eq. complexity_theorem, European call)

